

HW 13 - Series

Prob 1: Infinite Series: $\sum_{k=k_0}^{\infty} a_k$

(i) $k_0 = 0$ or $k_0 = 1$
 $a_k = \left(\frac{1}{2}\right)^{k+1}$ or $a_k = \left(\frac{1}{2}\right)^k$

(ii) $k_0 = 0$ or $k_0 = 1$
 $a_k = -3\left(\frac{1}{2}\right)^{k+1}$ or $a_k = -3\left(\frac{1}{2}\right)^k$

(iii) $k_0 = 1$

$$a_k = \frac{k}{(k+1)^2} - \frac{k-1}{k^2}$$

$$\begin{aligned} \sum_{k=1}^N a_k &= \left(\frac{1}{2^2} - \frac{0}{1^2}\right) + \left(\frac{2}{3^2} - \frac{1}{2^2}\right) + \left(\frac{3}{4^2} - \frac{2}{3^2}\right) + \dots + \left(\frac{N}{(N+1)^2} - \frac{N-1}{N^2}\right) \\ &= 0 + \frac{N}{(N+1)^2} \end{aligned}$$

$$\lim_{N \rightarrow \infty} \sum_{k=1}^N a_k = \lim_{N \rightarrow \infty} \frac{N}{(N+1)^2} = 0 \quad \checkmark$$

Prob 2: $a_n = \left(\frac{3}{4}\right)^n$, $b_n = \left(\frac{1}{4}\right)^n$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{a_n}{b_n} = \sum_{n=1}^{\infty} 3^n \quad \text{diverges}$$

Prob 3: $\sum a_n$ converges $\Rightarrow a_n \rightarrow 0^+$

$$\Rightarrow \frac{1}{a_n} \rightarrow \infty \Rightarrow \sum \frac{1}{a_n} \text{ diverges.}$$

Prob 4: Claim: $\sum a_n + b_n$ diverges

$$\sum a_n = A$$

$$\lim_{N \rightarrow \infty} \left(\sum_{n=1}^N a_n + \sum_{n=1}^N b_n \right) = \lim_{N \rightarrow \infty} \left(\underbrace{\sum_{n=1}^N a_n}_A + \underbrace{\sum_{n=1}^N b_n}_{\text{does not approach a finite \#}} \right)$$

$\searrow$ $\nearrow$
 does not approach a finite #.

Prob 5 : Since $1 + e^b + e^{2b} + e^{3b} + \dots = \sum_{k=0}^{\infty} (e^b)^k$ (Geometric)

This converges if $-1 < e^b < 1$ and has the sum:

$$\frac{1}{1 - e^b} = 9 \Leftrightarrow 1 = 9(1 - e^b)$$

$$\Leftrightarrow 1 = 9 - 9e^b$$

$$\Leftrightarrow +\frac{8}{9} = e^b \Leftrightarrow$$

$$\boxed{b = \ln\left(\frac{8}{9}\right)}$$

which satisfies $-1 < e^b < 1$ ✓

Prob 6 : $\underbrace{1 + 2r}_{a_1} + \underbrace{r^2 + 2r^3}_{a_2} + \underbrace{r^4 + 2r^5}_{a_3} + r^6 + \dots \quad (*)$

$$\Rightarrow a_k = r^{2k} + 2r^{2k+1}$$

$$\Rightarrow (*) = \sum_{k=0}^{\infty} r^{2k} + 2r \sum_{k=0}^{\infty} (r^2)^k = \sum_{k=0}^{\infty} (r^2)^k + 2r \sum_{k=0}^{\infty} (r^2)^k$$

$$= \frac{1}{1 - r^2} + \frac{2r}{1 - r^2}$$

$$= \frac{1 + 2r}{1 - r^2}$$

converges if

$$|r^2| < 1$$

or

$$|r| < 1$$